

Name of the experiment: Verification of half subtractor circuit using X-OR, NOT and AND.

Objective: The objective of this experiment is a half adder can be implemented by using X-OR, NOT and AND gate.

Theory:

Half Subtractor: It is a combinational logic circuit with two inputs and two outputs. Half subtractor is designed to subtract two binary single bit digit. It has two inputs and two outputs to indicate if a 1 has borrow.

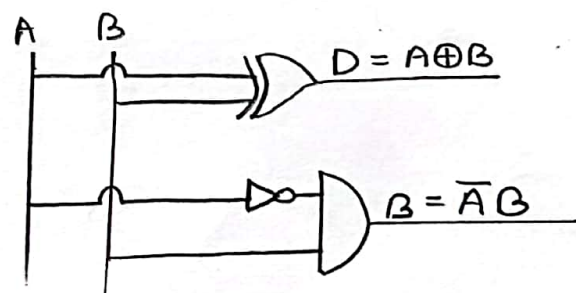
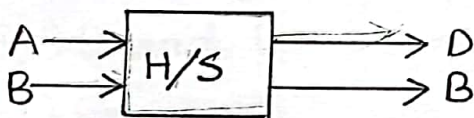


Fig: Half subtractor using X-OR, NOT and AND.

Truth Table :

Input		Output	
A	B	Difference	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

$$D = A \oplus B$$

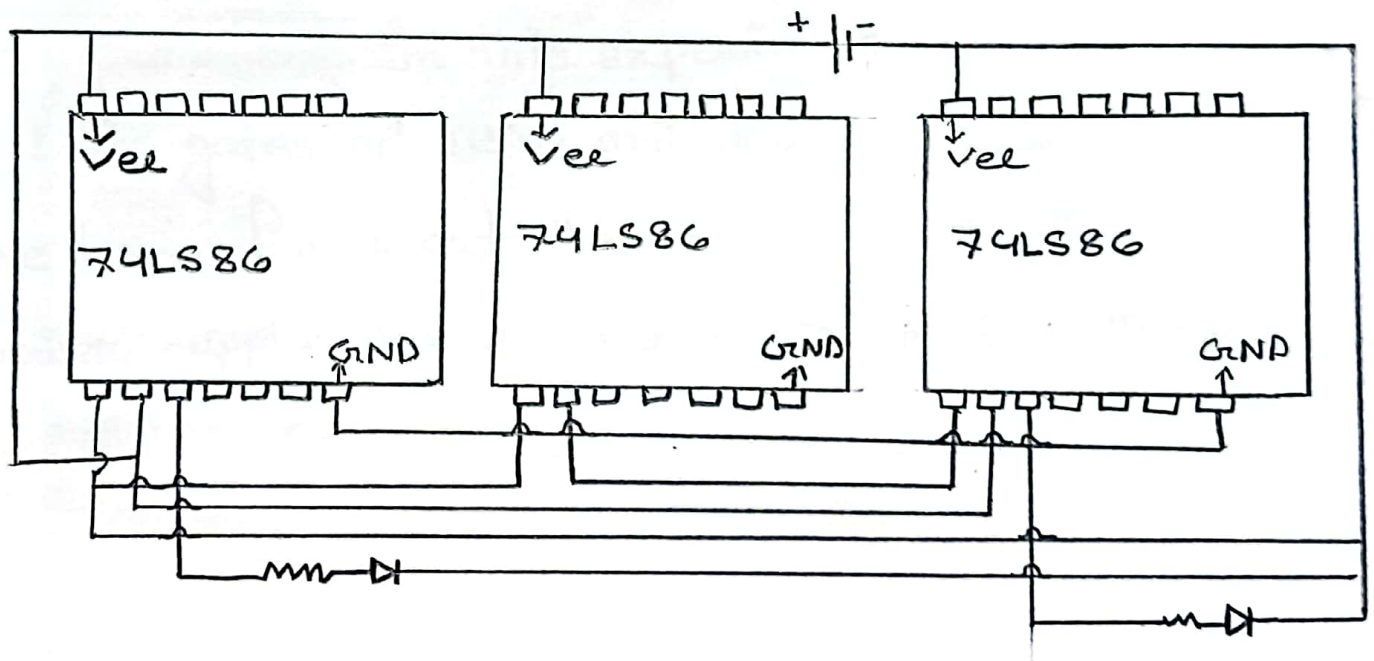
$$B = \bar{A}B$$

Apparatus :

- i) NOT gate.
- ii) AND gate.
- iii) X-OR gate.
- iv) Bread Board.
- v) Resistor
- vi) Wire
- vii) LED.

Procedure :

At first we have to identify the pin number of IC and then set it on the bread board connect the wire in the input pin and output pin connected to the resistor and then to the light. All the Vee and GND connected with the power supply.



Result :

A	B	Difference	LED	Borrow	LED
0	0	0	OFF	0	OFF
0	1	1	ON	1	ON
1	0	1	ON	0	OFF
1	1	0	OFF	0	OFF

Discussion :

In this experiment, we have checked all the gates of IC's and found the active gates. We have found out the position that IC's have power supply. We connected Vee and GND correctly.

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